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M 420 FIRE DETECTION AND PROTECTION SYSTEM

1 GENERAL

This section of the specification describes the technical requirements for the Fire Fighting System and covers the minimum requirements for design, engineering, manufacture, erection and inspection and testing. The sections of this specification entitled M-001, M-002, M-004, M-005, M-006 also include technical requirements that are applicable to the Fire Fighting System covered by this section of the specification. In the event of any conflict between the project requirements and the requirements of this section the latter shall have precedence.

The Fire Fighting System consists of Fire Protection System and Fire Detection System.

The Bidder shall provide three (3) Fire Detection and Protection system for Block 2, 3 and 4 Muara Tawar Add-on Project requirement. One (1) Fire Detection and Protection System shall be installed on each block.

The Fire Protection System equipment shall be complete in every respect and suitable for safe and reliable operation.

This specification does not enumerate or describe all the materials and equipment to be supplied and all the services to be performed. All materials and equipment shall be provided as are required to make a complete, properly functioning installation and shall conform to the highest standards of engineering design and workmanship.

The firefighting system shall be in accordance with the NFPA recommendations.

1.1 Fire water demand calculation

The fire water demand of the power plant shall be calculated and stated in the offer considering a worst case scenario based on NFPA 850. A safety factor of approximately 1.15 shall be added.

2 SCOPE OF WORK

An onsite fire protection system shall be provided in accordance with Part 4, Subsection 1.2 “Summary of Work” and as described herein.

This clause sets out the scope of equipment and services as well as specially requested supplies and services but does not exclude other necessary components and services not specifically mentioned herein.

The Bidder shall provide and supply Fire Protection System equipment complete including all necessary components, spare parts and services for the Muara Tawar combined cycle plant.

The scope of supply and services shall include the design, manufacture, construction, factory testing, packing for transport, transport and delivery to site, unloading at site, storing, complete erection, testing at site, painting, commissioning, putting into normal operation, tie-in with existing facilities, reliability and performance testing of firefighting system and associated equipment including but not limited to the following:

Sub systems:

- water distribution systems as extension of the existing water distribution systems
- sprinkler and spray water systems
- Clean Agent Extinguisher systems for Steam Turbines Block 2, 3 & 4.
- aboveground hydrants and hose reels
- foam systems
- portable and mobile extinguishers

Components and others:

- pipework and valves, anchor bolts and supports
- extension pipeline from the existing fire protection system including all of necessary valves, control, instrument, electrical, pipe support, etc.
- all supply and services at tie-in points
- control and monitoring,
- complete detailed labeling of all installations,

- consumables such as first fill and any additional required,
- set of special tools and equipment for maintenance, inspection and repair,
- all standard equipment and accessories which are normally included in the supply schedule but which are not separately listed,
- detail design and calculation including influence on existing firefighting installations .
- participation in all implementation discussions,
- compiling and local collection of required data,
- training, inspection and supervision services by the Manufacturer
- spare parts and wear-and-tear parts acc. to manufacturer's recommendation
- complete documentation as specified.

The fire protection system shall generally follow NFPA 850.

The following main areas are considered to be protected:

Building or Area	Fire fighting System
Steam turbine lube oil package, lube oil piping	Automatic sprinkler
Steam turbine bearings	Automatic Sprinkler
Steam turbine building indoor	Wet stand pipe house system
Main and auxiliary transformer	Automatic water spray
Control room	Total flooding inert gas
Electrical equipment room	Total flooding inert gas
Yard	Hydrants strategically placed 60 m apart
Common	Protective signaling for fire detection systems with main panel in the control room. This equipment will include: • one central supervisory control panel to monitor the status of zones, with visual indications, audible alarm and test provisions • local fire alarm panels and zone/terminal boxes to interconnect the facility fire protection systems as required.

Building or Area	Fire fighting System
	• area fire/smoke detectors where required for automatic suppression system actuation • manual pull stations fire alarm horns (audible throughout the site).

Extinguishers shall be sized, rated and spaced in accordance with NFPA 10. Local buildings fire alarms, automatic fire detectors and the fire signaling panel shall be in accordance with NFPA 72.

It shall be assured that a dedicated two (2) hour fire water supply to cover the system design flow rate is available for the facility in accordance with NFPA.

Existing main fire header shall be extended to serve strategically placed (approximately 60 m apart) yard hydrants and supply water to the sprinkler and spray system.

The fire water distribution system shall incorporate sectionalizing valves so that a failure in any part of the system can be isolated while allowing the remainder of the system to function properly.

Portable fire extinguisher shall be provided around the power plant, and all building.

Miscellaneous protective clothing helmets, tools, breathing apparatus, etc. as specified.

The fire protection system shall include but not be limited to the system as described herein. All systems shall be furnished and erected complete in every respect, from point of connection with supply lines complete to each protection area and shall include all appurtenances and accessories, such as nozzles, piping, piping manifolds and tubing, piping hangers and supporting structures, automatic and manual controls, alarms, smoke, temperature and flame detecting devices as specified or required, pressure gauges switches, flow switches, air and water, control panels, service water pumps, emergency water pumps, motor valves, pressure tanks diesel engine etc.

3 CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.

The Codes and Standards extracted from Part 4, Subsection 2.5 “Codes and Standards” or other equivalent international standards are specifically applicable to the design, manufacturer and testing of the equipment included in this specification.

ANSI	American National Standards Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society for testing and Material
NFPA	National Fire Protection Association
UL	Underwriter Laboratories Inc
FM	Factory Mutual Insurance Co

4 DESIGN AND OPERATING CONDITIONS

Each fire protection system design and installation shall be furnished with all necessary equipment to form a complete system as defined by the National Fire Protection Association (NFPA) codes, standard and recommended practices, codes and this specification.

Yard mains and outdoor fire hydrants shall be in accordance with NFPA 24 and 26.

Stand pipes and hose systems shall be per NFPA 14.

Automatic fire detectors, Fire Alarms and fire protection actuation system, shall be per NFPA 72A-E

Portable fire extinguisher and dry chemical extinguishing system shall be per NFPA 10 and 17.

Automatic and manual sprinkler systems shall be per NFPA 13, 231 and 231C.

Protection of electronic computer/data processing equipment shall be per NFPA 2001

Fire doors shall be per NFPA 80

Water spray fixed systems shall be per NFPA 15.

Fire water valves shall be per NFPA 13, 14, 15, 20, 24 and 26.

The water monitors shall be provided per NFPA 11.

Protective clothing and helmets shall be provided per NFPA 1971 and 1972.

Breathing apparatus shall be provided per NFPA 1981.

All equipment and materials used in the fire detection and protection systems shall meet the approval of recognized testing agency, Factory Mutual Insurance Corp. or its international equivalent and have a Indonesia Board of Standards and Appeals number.

The design of the fire detection and protection system shall be adequate for the Site conditions as specified.

4.1.1 Fire Fighting Pump

It shall be assured that a dedicated two (2) hour fire water supply to cover the system design flow rate is available for the facility in accordance with NFPA.

The firefighting water source shall be pumped from raw water tank.

The Firefighting pump house being the main supply will consist of:

- One (1) x 100% electric jockey pump
- One (1) x 100% electric driven main pump
- One (1) x 100% diesel engine driven main pump
- One (1) expansion tank arrangement

The water demand and required pressure for the worst case condition shall be ensured by the electric driven main pump, the second duty diesel engine driven pump shall be stand-by, for the case of mains supply failure.

The HSD fuel oil tanks and dike shall be furnished with foam firefighting systems.

4.1.2 Clean Agent Fire Extinguisher System

Clean agent tank with accessories shall be installed in the new combined cycle plant. The Bidder shall design the entire new system fire detection and protection for control cubicles, control room, computer room and other location proposed by the Bidder.

4.1.3 Fire Protection Panel

The central fire protection panel shall be designed by the Bidder so as to cover the fire protection and fire alarm system for the entire plant, and shall be installed in the central Control Room.

The size of the fire protection panel shall be designed to suit the control room. The central fire protection panel shall consist of graphic panel, control unit for each pump, fire alarm, operation for remote operating valves, and others as shown below.

(a) Graphic panel

The outline of the Combined Cycle Power Plant shall be shown on the graphic panel to indicate the fire detecting points, the operating of pumps, and the opening of remote valves and motor valves.

(b) Fire water pump control unit

5 DETAILED REQUIREMENTS

5.1 General Requirements

The Power Plant shall be designed and built with provision of safe operating environment for equipment and personnel. This shall be achieved by separation and segregation of equipment with sufficient distances and by selection of suitable equipment and materials.

Hazardous areas shall be identified and suitable equipment shall be selected for use in such areas. Different firefighting systems shall be adopted according to the operational characteristics of the plant areas and buildings to be protected.

An automatic fire detection and alarm system shall be provided, with detection devices selected to suit particular risks, and with a control system designed to provide operating and fire brigade staff with sufficient information to identify and respond correctly to any fire detected.

Automatic and manual extinguishing systems shall be provided to limit the consequences of fires in major items of plant and minimize resulting reductions in generating capacity.

Water based fire systems shall be extended from existing firefighting systems tie-in. The system shall be provided with sectional isolating valves to allow the maximum supply in the event of a fault. Hydrants shall be provided at strategic locations, with hoses and nozzles kept in suitable cabinets.

For all transformers an automatic spray water system shall be supplied.

Portable fire extinguishers shall be provided throughout the site, with larger capacity units sited at increased risk areas.

Hose reels and internal hydrants shall be provided within buildings where appropriate.

All the equipment and escape routes shall be clearly marked.

For fire protection and detection purposes the overall site shall be divided into separate fire zones and the alarms and control requirements for each zone shall be actioned through the central control and alarm panel.

The firefighting system shall be in accordance with the NFPA recommendations.

In the case that the firefighting system is in operation the required delivery and pressure for the worst condition are ensured by a main duty electric pump; a second duty pump, diesel engine driven, is stand-by, for the case of mains supply failure.

A supply will be provided from the existing plant by opening manually the interconnection valve of the fire water fighting system to the existing system.

The water distribution system is an underground system and comprises different circuits. Shut-off valves shall be foreseen to isolate parts of the system, if and when required; these valves have to be normally kept open.

The Work shall include the complete design, furnishing and delivery installation and testing of a complete fire protection system.

5.1.1 Automatic Operation

Upon detection of a fire by specified fire detectors, the Water supply valve for the system shall automatically open to release water to the piping.

5.1.2 Manual Operation

- a. In addition to automatic operation, each detector circuit system shall be furnished with a minimum of one break glass station for manual operation of the system. Also, each automatic water supply valve shall also be provided with means for local mechanical operation.
- b. At the time that the automatic water supply valve is opened, an alarm shall be given by a fire bell which shall be audible throughout the zone covered. A second independent signal shall be provided for the Central Control Room Annunciator panel, furnished by Bidder. A DPDT (Double Pole Double Throw) contact activated by a water flow pressure switch shall be provided by the bidder to start the Equipment.
- c. Each automatic water supply valve shall be designed so as not to activate on momentary pressure surge in the system.

5.1.3 Water Manual Valve

Each fire zone sprinkler header will contain a lock type manual gate valve with a tamper switch complete with position indicator and locking device, to allow Owner to lock valve in the open position, as recommended by Bidder and as approved. Sprinkler orifice diameter shall be sized to avoid the need of water strainers.

5.1.4 Fire Detection and Alarm Systems

The overall project shall be provided with fire alarm systems in accordance with the design requirements.

The primary purpose of the system is to give an early warning of dangerous conditions to occupants enabling them to evacuate the building or area safely.

A combined manual and automatic Fire Detection and Alarm System shall be provided in all fire zones.

5.1.4.1 Break Glass

Break Glass Alarms shall be provided at all exits and stairwells and at location not greater than 50 meters apart, for all buildings.

The break glass shall be of robust construction and shall be coloured signal red. A striker shall be provided adjacent to all break glass to facilitate breaking the glass.

5.1.4.2 Audible Alarms

Bidder shall install and locate audible alarms in convenient locations and submit his design to the Owner for review.

Alarm bells shall be installed in all office areas and shall have coated metal gongs or domes with fully enclosed dustproof mechanism.

Klaxons or sirens shall be installed in plant locations and shall be of weather proof type with sheet metal enclosures.

5.1.4.3 Detectors

In addition to the detection equipment required for the automatic fire protection system. The Bidder shall provide conveniently located fire alarm detectors and submit his design to the Owner for review. The detector system shall be designed, furnished and installed according to the requirements of NFPA 72D and 72E.

5.1.5 Sprinklers

Each system, connected to the water distribution, network starts delivering water automatically when the high temperature causes breaking of the sprinkler bulb and stops by closing the relevant valve.

5.1.6 Deluge System

Each oil-filled transformer shall be provided with all equipment necessary to form a complete automatic water spray deluge system classification set forth in NFPA standard including but not limited to local control switches, pipe fitting, automatic water supply valves hydraulically operated deluge valves), closed type sprinkler nozzles as heat detectors, drain valves, fire protection devices, alarm sounder, open head type sprinklers, controls and instrumentation, specialties, hangers and supports, wiring and conduit.

5.2 Yard Hydrant System/Water Hose Station

The Bidder shall furnish a complete system of hydrants and fire hoses in accordance with applicable NFPA Standards. The system shall include but not be limited to the following:

Class B (as defined in NFPA 291) hydrants shall be provided.

Each hydrant shall be fitted with an isolating valve in the connection from the main.

The fire hoses shall be provided in accordance with the latest requirement of NFPA 24 Chapter 5.

Each of the fire hoses shall be provided with instantaneous coupling with male and female coupling at opposite ends.

Hydrant boxes (hose house) shall be provided adjacent to each hydrant in accordance with NFPA 24, Chapter 5.

5.3 Clean Agent Extinguishing System

The Bidder shall furnish the Clean Agent Extinguishing System design for review by the Owner.

5.4 Portable Fire Extinguisher

Multipurpose dry chemical and carbon dioxide extinguisher shall be provided for all locations proposed by Bidder for review by the Owner.

The portable fire extinguisher shall be complete with wall mounted fixtures.

Multipurpose dry chemical shall be of 8.8 kg (approx) capacity.

Where extinguishers are provided externally, or in other areas where they may be subjected to the weather, they shall be contained inside a protective cabinet.

The extinguishers shall of the type operated by means of a lever provided with a safety pin, which allows for the partial discharge.

5.5 Protective Clothing and Helmets

The Bidder shall provide:

- 10 pcs Fire Protection suits complete with helmet, suit, gloves, shoes.
- Two fire rescue suit complete with helmet, suit, gloves and shoes shall be provided in a fire rescue suit box. The box shall be provided by the Bidder. This box will be installed at the fire pump house within a separate room.

5.6 Breathing Apparatus

The apparatus shall be provided with:

- Pressure regulation.
- Pressure reduction for the convenience of the user.
- Bell alarm for oxygen indicator.
- Diaphragm at the masked head.

The dials shall be such that they can be read in the dark.

5.7 Fire stopping

Where cables, pipes or air duct work pass through fire compartment barriers any space left in passageways for services shall be closed to ensure a complete stop to fire, having fire resistance at least equal to that of the barrier itself.

All fire stopping methods employed shall allow for the movement of the services to which they are associated (vibration or thermal movement) and resist misuse or abuse while remaining effective as a barrier to heat, smoke and other products of combustion.

The use of brittle materials such as plaster or cement mortar shall not be acceptable. The Bidder shall show that the fire stops provided are completely suitable and will remain effective for the extended periods without maintenance or supervision.

5.8 Fire retardant coatings

In order to prevent the spread of fire by cables in floor voids and other enclosed space not protected with fire extinguishing systems, such cables shall be completely coated with a water resistant fire retardant coating which will not affect the current rating of the cables.

The coating shall be suitable for application by brush to provide a minimum of ten minutes fire resistance protection when tested in accordance with ISO 834 and ASTM E 119.

The fire retardant coating shall prevent the propagation of fire along cables for a minimum of sixty minutes when the cables are on vertical trays when a furnace of 800°C is placed beneath them. The Bidder shall submit test results to show the effectiveness of the fire retardant coating.

The coating shall be weather resistant, shall not change its fire resistance with age and shall not use solvents which are toxic or provide a fire hazard during application.

When dry it shall be flexible and tough with good impact resistance but have poor tear strength in order to allow for the removal and addition of cables.

5.9 Fire dampers

Where air ductwork passes through a fire compartment barrier it shall contain a fire damper.

Air ductwork shall be strengthened at fire compartment barriers to resist buckling and shall be securely fixed to the barrier to prevent the spread of fire around the air duct. Within the duct a heat actuated multi or single blade fire

damper which shall provide a resistance to fire at least equal to the fire compartment barrier.

Smoke detectors shall also be provided to detect smoke within the air ductwork, initiate a level 1 alarm of fire and close all the fire dampers associated with the fire compartment in which smoke is detected.

5.10 Escape routes

Adequate escape routes shall be provided for all personnel working in the power station building; in accordance with the appropriate codes and standards.

The escape routes shall be clearly marked and provided with emergency lighting. They shall be routed away from hazardous areas and protected, floors and doors to allow safe escape for personnel, and safe access for firefighting personnel. Where possible, more than one significantly different escape route shall be provided. Staircases on escape routes shall, where appropriate, be provided with smoke vents.

The Bidder shall provide the drawings depicting Evacuation Routes/Escape Routes and Fire Exit Doors for all areas, all floors and all buildings under the contract. These drawings shall be displayed in appropriate locations.

6 INSPECTION, TESTING AND GUARANTEES

The Bidder shall test and place in successful operation all equipment and materials furnished by him. The testing of equipment shall include testing of entire system for proper operation of equipment, devices and wiring furnished and/or installed of the complete fire protection system.

After completion of the installation, Bidder shall provide certificate of completion if any additional piping, spray nozzles, pilot sprinklers, etc, may be required to provide a system satisfactory to Owner, or Owner's insurance underwriter, these shall be provided by Bidder at no extra cost to Owner.

After successful placement in operation of the complete fire protection system, the Bidder shall submit a fire protection system report.

All instruments used for the purpose of carrying out reliability, function or certification tests must have an in-date test/calibration certificate. All instru-

ments must be recalibrated and certified in the time period regulated by the relevant standards, but not less than every six months.

6.1 Performance tests

Each firefighting pump as other firefighting equipment shall be Performance tested to demonstrate that the operation complies with the Guaranteed Performance and to provide base data for the assessment of Performance throughout the operating life of the equipment. The extent of and procedures for performance tests shall be to the approval of the Owner.

Unless otherwise specified in the Equipment Requirements the programme of performance test shall include at least the following:

- determination of maximum continuous rating;
- determination of normally operation rating data and performances.;
- leak test.

7 QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality Assurance/Quality Control shall conform to the requirements of Part 4, Technical Requirements, Subsection 2.7.

8 ELECTRICAL AND CONTROL REQUIREMENTS

All detectors shall be wired in loops containing end of line resistor so that the detector circuit continuity is continuously monitored and the circuit troubles are automatically annunciated.

Each fire zone with detectors used to active fire suppression equipment shall be equipped with a local Fire Alarm Control Panels (FACP). As deemed appropriate, the local FACP's will serve the designated fire zones on either an individual zone or a multiple zone basis.

The FACP's shall provide local control, monitoring and audible/visual indication for the zone fire suppression and detection equipment, allowing personnel in the area to react to alarm conditions. In addition, the FACP's shall relay zone fire, trouble and supervisory alarms to the Central Control Room initiating responsive operator action through audible alarms, visual indication on the annunciator and the main control board.

Local Fire Alarms Control Panels (FACP's) shall be installed nearby but outside fire protection and detection zones and, each zone, shall be equipped with indicating lights for the following:

- Power Failure
- Open circuit or ground in any detector, manual station, or solenoid valve circuit
- Isolation or shutoff valve not fully open
- Fire condition detected by fire detector, manual station, or water pressure or flow switches.
- Pre-alarm condition when thermistor cable is used.

The local Fire Alarm Control Panels shall also generate the necessary output signals to alarm the fire and circuit trouble in its zone(s) of protection locally and at the Main Control Room for Central Supervisory function. They shall also provide the output signals for automatic fire suppression and to shut down plant or VAC equipment, if required.

The wiring from detector circuits, manual stations, damper, pressure and flow switches; solenoid valves, etc. shall be monitored for continuity and grounds by the local control panel. Upon detection of a single open or ground conductor, a trouble signal shall also be transmitted to the Central Control Room annunciator system.

All FACP's shall have two source of power - primary and secondary. The primary power shall be monitored and alarmed upon failure.

The secondary power using storage batteries in the FACP's supply shall be capable of supplying the energy to the system under the maximum normal load for 24 hours and then be capable of receiving one fire alarms signal persisting for five minutes in the event of total failure of low voltage of the primary power supply.

The secondary power supply shall automatically transfer to operate the system within 30 seconds of the loss of the primary power supply.

The FP detector circuits, FACP's and associated wiring shall meet the applicable requirements of Technical Specifications and in accordance with NFPA Standard, It is the responsibility of the Bidder to select the appropriate fire de-

tection devices in accordance with the NFPA standard as applicable in each area or zone.

The detailed description of panel shall be referred to Part 4, Technical specification I-002.

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